

National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 01 May 2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई -400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापलंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु -560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 30.04.2024.

महोदय/Dear Sir,

आईईजीसी-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 30-अप्रैल-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 30 April 2024, is available at the NLDC website.

धन्यवाद,

Report for previous day

Date of Reporting: 01-May-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	61837	66392	54581	28957	2789	214556
Peak Shortage (MW)	0	0	0	350	0	350
Energy Met (MU)	1357	1635	1393	667	56	5108
Hydro Gen (MU)	199	55	56	30	14	354
Wind Gen (MU)	23	134	47	-	-	203
Solar Gen (MU)*	184.20	102.85	134.63	3.62	1.18	426
Energy Shortage (MU)	0.44	0.00	0.00	3.77	0.00	4.27
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	62789	73936	66470	31895	3059	224181
Time Of Maximum Demand Met	19:49	15:31	14:55	23:16	18:54	22:42

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.035	0.00	0.00	1.70	1.70	71.09	27.21

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	8231	0	151.0	38.2	-1.8	430	0.00
	Haryana	8784	0	172.6	98.9	-1.0	243	0.00
	Rajasthan	14110	0	291.1	69.3	-1.1	316	0.36
	Delhi	5016	0	105.5	87.9	-0.6	229	0.00
	UP	24958	0	502.0	217.8	0.2	780	0.00
	Uttarakhand	2294	0	47.9	25.9	-0.3	235	0.08
	HP	1649	0	31.1	13.5	-0.2	61	0.00
	J&K(UT) & Ladakh(UT)	2370	0	47.6	26.7	-1.4	310	0.00
	Chandigarh	246	0	4.6	5.4	-0.7	16	0.00
WR	Railways_NR ISTS	188	0	3.9	3.6	0.2	82	0.00
	Chhattisgarh	6097	0	136.8	75.5	-1.7	290	0.00
	Gujarat	23010	0	478.6	148.2	-5.9	535	0.00
	MP	13151	0	292.4	174.8	-2.6	664	0.00
	Maharashtra	28703	0	648.3	217.8	-3.9	855	0.00
	Goa	783	0	17.3	12.5	4.4	133	0.00
	DNHDDPDCL	1250	0	29.3	29.3	0.0	47	0.00
	AMNSIL	893	0	20.2	7.2	-0.1	267	0.00
SR	BALCO	523	0	12.5	12.6	-0.1	21	0.00
	Andhra Pradesh	13418	0	255.5	102.4	-0.4	959	0.00
	Telangana	11224	0	232.8	105.1	-0.4	912	0.00
	Karnataka	16693	0	336.4	158.2	1.0	1219	0.00
	Kerala	5476	0	113.3	86.9	2.3	420	0.06
	Tamil Nadu	20701	0	443.6	277.2	-3.1	804	0.00
	Puducherry	514	0	11.5	11.2	-0.3	30	0.00
ER	Bihar	7051	0	138.5	125.5	1.8	831	2.55
	DVC	3551	0	75.5	-47.0	0.3	347	0.00
	Jharkhand	2187	0	47.3	32.5	0.7	174	1.22
	Odisha	6860	0	140.1	61.6	1.5	512	0.00
	West Bengal	12756	0	264.0	114.5	-0.2	380	0.00
	Sikkim	97	0	1.4	1.4	0.0	52	0.00
NER	Railways_ER ISTS	15	0	0.1	0.2	-0.1	12	0.00
	Arunachal Pradesh	184	0	2.6	2.6	-0.1	43	0.00
	Assam	1826	0	34.7	28.6	-0.1	121	0.00
	Manipur	207	0	3.0	3.0	0.0	21	0.00
	Meghalaya	309	0	5.2	3.7	-0.2	96	0.00
	Mizoram	127	0	2.0	1.8	-0.1	33	0.00
	Nagaland	145	0	2.3	2.4	0.0	7	0.00
	Tripura	307	0	5.9	5.5	-0.3	87	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	-2.3	-13.6	-22.8	-30.5
Day Peak (MW)	-280.0	-805.0	-1049.0	-1455.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	92.5	-341.3	241.1	0.1	7.6	0.0
Actual(MU)	76.1	-358.7	257.4	7.9	9.0	-8.3
O/D/U/D(MU)	-16.4	-17.4	16.2	7.8	1.5	-8.3

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	1958	3490	2368	4121	647	12584	43
State Sector	5101	6796	2578	2054	89	16617	57
Total	7059	10286	4946	6175	736	29201	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	852	1679	801	718	15	4064	73
Lignite	26	15	71	0	0	112	2
Hydro	199	55	56	30	14	354	6
Nuclear	31	52	75	0	0	158	3
Gas, Naptha & Diesel	45	94	7	0	23	169	3
RES (Wind, Solar, Biomass & Others)	216	238	212	5	1	673	12
Total	1369	2133	1222	753	54	5531	100

Share of RES in total generation (%)	15.80	11.17	17.36	0.62	2.20	12.16
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	32.59	16.16	28.13	4.65	28.94	21.43

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.062
Based on State Max Demands	1.096

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	224052	15:30	0
Non-Solar hr	224181	22:42	93

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 07:00 to 17:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 01-May-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	48	0.0	1.3	-1.3
3	765 kV	GAYA-VARANASI	2	971	240	5.7	0.0	5.7
4	765 kV	SASARAM-FATEHPUR	1	276	267	0.0	1.7	-1.7
5	765 kV	GAYA-BALIA	1	84	451	0.0	6.6	-6.6
6	400 kV	PUSAULI-VARANASI	1	0	126	0.0	1.9	-1.9
7	400 kV	PUSAULI -ALLAHABAD	1	79	0	0.8	0.0	0.8
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	787	197	5.4	0.0	5.4
9	400 kV	PATNA-BALIA	2	378	614	0.0	3.6	-3.6
10	400 kV	NAUBATPUR-BALIA	2	211	179	0.0	0.3	-0.3
11	400 kV	BIHARSHARIFF-BALIA	2	637	0	7.1	0.0	7.1
12	400 kV	MOTIHARI-GORAKHPUR	2	391	298	0.0	0.6	-0.6
13	400 kV	BIHARSHARIFF-VARANASI	2	0	0	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	54	193	0.0	0.8	-0.8
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.6	0.0	0.6
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAUJI	1	0	0	0.0	0.0	0.0
ER-NR						19.7	16.7	3.0
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	2287	0	35.6	0.0	35.6
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	1422	0	20.2	0.0	20.2
3	765 kV	JHARSUGUDA-DURG	2	108	403	0.0	5.3	-5.3
4	400 kV	JHARSUGUDA-RAIGARH	4	62	314	0.0	3.5	-3.5
5	400 kV	RANCHI-SIPAT	2	290	27	3.4	0.0	3.4
6	220 kV	BUDHIPADAR-RAIGARH	1	34	89	0.0	0.8	-0.8
7	220 kV	BUDHIPADAR-KORBA	2	92	17	0.6	0.0	0.6
ER-WR						59.7	9.6	50.2
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	521	0.0	12.4	-12.4
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1505	36.2	36.2	0.0
3	765 kV	ANGUL-SRIKAKULAM	2	0	2681	0.0	49.9	-49.9
4	400 kV	TALCHER-I/C	2	427	192	4.8	0.0	4.8
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						36.2	98.4	-62.2
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	491	30	2.6	0.0	2.6
2	400 kV	ALIPURDUAR-BONGAIGAON	2	850	0	7.8	0.0	7.8
3	220 kV	ALIPURDUAR-SALAKATI	2	135	0	1.4	0.0	1.4
ER-NER						11.8	0.0	11.8
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	1416	0	22.1	0.0	22.1
NER-NR						22.1	0.0	22.1
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	2001	0.0	42.8	-42.8
2	HVDC	VINDHYACHAL B/B	-	758	0	12.1	0.0	12.1
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1911	0.0	39.9	-39.9
4	765 kV	GWALIOR-AGRA	2	184	1532	0.3	18.0	-17.8
5	765 kV	GWALIOR-PHAGI	2	1435	1038	7.4	5.2	2.2
6	765 kV	JABALPUR-ORAI	2	406	583	0.0	9.1	-9.1
7	765 kV	GWALIOR-ORAI	1	524	33	6.0	0.0	5.9
8	765 kV	SATNA-ORAI	1	0	803	0.0	15.5	-15.5
9	765 kV	BANASKANTHA-CHITORGARH	2	2217	851	14.2	4.1	10.1
10	765 kV	VINDHYACHAL-VARANASI	2	0	2936	0.0	53.4	-53.4
11	400 kV	ZERDA-KANKROLI	1	433	93	3.6	0.2	3.3
12	400 kV	ZERDA -BHINMAL	1	432	93	3.6	0.2	3.3
13	400 kV	VINDHYACHAL -RIHAND	1	966	0	22.0	0.0	22.0
14	400 kV	RAPP-SHUJALPUR	2	877	121	7.8	0.1	7.7
15	400 kV	NEEMUCH-Chittorgarh	2	994	20	9.0	0.0	9.0
16	220 kV	BHANPURA-RANPUR	1	0	120	0.0	2.3	-2.3
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.1	-2.1
18	220 kV	MEHGAON-AURAIYA	1	100	0	1.4	0.0	1.4
19	220 kV	MALANPUR-AURAIYA	1	70	0	0.9	0.0	0.9
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						88.1	193.0	-104.9
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1008	0.0	18.9	-18.9
2	HVDC	RAIGARH-PUGALUR	2	0	4010	0.0	92.0	-92.0
3	765 kV	SOLAPUR-RAICHUR	2	0	1253	0.0	21.6	-21.6
4	765 kV	WARDHA-NIZAMABAD	2	0	2156	0.0	38.8	-38.8
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2850	0.0	51.6	-51.6
6	400 kV	KOLHAPUR-KUDGI	2	969	0	15.5	0.0	15.5
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	82	1.8	0.0	1.8
9	220 kV	XELDEM-AMBEWADI	1	0	128	2.5	0.0	2.5
WR-SR						19.8	222.9	-203.1

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)	
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	256	72	168	4.03
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	-307	84	-102	-2.45
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-281	-99	-197	-4.72
	NER	132kV GELEPHU-SALAKATI	15	-14	4	0.10
	NER	132kV MOTANGA-RANGIA	45	0	33	0.79
NEPAL	NR	NEPAL IMPORT (FROM UP)	-77	0	-46	-1.10
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-77	0	-54	-1.29
	ER	NEPAL IMPORT (FROM BIHAR)	-325	-19	-152	-3.66
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-480	-53	-313	-7.51
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-945	-648	-856	-20.55
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1455	0	-1271	-30.51
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-104	0	-93	-2.24

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	2.27	0.00	0.00	1.04	0.00	0.00	3.31
Nepal	0.29	0.00	5.99	0.00	0.00	0.00	0.00	0.00	6.28
Bangladesh	20.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.53
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	20.82	0.00	8.26	0.00	0.00	1.04	0.00	0.00	30.12

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.32
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.32

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.32	0.00	-2.27	0.00	0.00	-1.04	0.00	0.00	-2.99
Nepal	-0.29	0.00	-5.99	0.00	0.00	0.00	0.00	0.00	-6.28
Bangladesh	-20.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-20.53
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-20.50	0.00	-8.26	0.00	0.00	-1.04	0.00	0.00	-29.80